Raw moments	Raw data ($d=x-A$)	Discrete data ($d=x-A$)	Continuous data ($d=(x-A)/c$)
μ_1'	$\frac{\sum d}{n}$	$\frac{\sum fd}{N}$	$\frac{\sum fd}{N}\times c$
μ_2'	$\frac{\sum d^2}{n}$	$\frac{\sum fd^2}{N}$	$\frac{\sum fd^2}{N}\times c^2$
μ_3'	$\frac{\sum d^3}{n}$	$\frac{\sum fd^3}{N}$	$\frac{\sum fd^3}{N}\times c^3$
μ_4'	$\frac{\sum d^4}{n}$	$\frac{\sum fd^4}{N}$	$\frac{\sum fd^4}{N}\times c^4$

6.6.2 Central Moments:

Central moments can be defined as the arithmetic mean of various powers of deviation taken from the mean of the distribution. The rth central moment is denoted by μ_r, $r = 1, 2, 3...$

Central moments	Raw data	Discrete data	Continuous data $d'=\frac{(x-\bar{x})}{c}$
μ_1	$\frac{\sum(x-\bar{x})}{n}=0$	$\frac{\sum f(x-\bar{x})}{N}=0$	$\frac{\sum fd'}{N}\times c$
μ_2	$\frac{\sum f(x-\bar{x})^2}{N}=0$	$\frac{\sum f(x-\bar{x})^2}{N}=\sigma^2$	$\frac{\sum fd'^2}{N}\times c^2$
μ_3	$\frac{\sum(x-\bar{x})^3}{n}$	$\frac{\sum f(x-\bar{x})^3}{N}$	$\frac{\sum fd'^3}{N}\times c^3$
μ_4	$\frac{\sum(x-\bar{x})^4}{n}$	$\frac{\sum f(x-\bar{x})^4}{N}$	$\frac{\sum fd'^4}{N}\times c^4$

In general, given n observation $x_1, x_2, \ldots, x_n$ the rth order raw moments ($r = 0,1,2,...$) are defined as follows:

$$\mu'_r = \frac{1}{N}\sum f(x-A)^r \text{ (about } A)$$

$$\mu'_r = \frac{\sum fx^r}{N} \text{ (about origin)}$$

$$\mu_r = \frac{1}{N}\sum f(x-\bar{x})^r \text{ (about mean)}$$

NOTE

Raw moments are denoted by μ'_r and central moments are denoted by μ_r

Ch6_11th Stats.indd 175 2/3/2019 11:02:17 AM